AKSHAYA INSTITUTE OF TECHNOLOGY, TUMAKURU
Approved by AICTE, New Delhi, Affiliated to VTU, Belgaum, Recognized by Govt. of Karnataka
Obalapura Post, Lingapura, Koratagere Road, Tumakuru - 572 106, Karnataka

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Academic year: 2023-24'

Notes for Module-1

"ELECTROMAGNETIC THEORY"

[BEC401]

Prepared by:
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AKSHAYA INSTITUTE OF TECHNOLOGY

Lingapura,Obalapura Post,Koratagere Road,Tumakuru

DEPARTMENT OF CIVIL ENGINEERING

VISION

To achieve excellent standards of quality education by keeping pace with rapidly changing technologies and to create technical man power of global standards in civil engineering with capabilities of accepting new challenges.

MISSION

1. To impart quality education in civil engineering to meet the requirements of all stake holders.
2. To serve society and nation by providing professional skills in civil engineering.
3. To create competent professionals in design and development of civil engineering systems and contribute towards research and development activities.

Program Specific Outcomes (PSOs)

After successful completion of Civil Engineering Program students will be able to

1. Utilize multidisciplinary knowledge required for satisfying industry / global needs.
2. Capable to design and build civil engineering based systems in the context of environmental, economical, and societal requirements.
3. Able to apply professional skills in the areas of Civil Engineering such as Structural Engineering, Environmental Engineering, Geotechnical Engineering, Water Resources Engineering and Transportation Engineering.

Program Educational Objectives (PEOs)

1. Graduates excel in the fundamentals of mathematics, basic science, communi-cation and Civil Engineering concepts to solve the real world Civil Engineering problems by imparting technical knowledge.
2. Graduates capable to analyze, design, construct, maintain and rehabilitate all civil engineering structures without compromising on monetary, environmental, ecological and social responsibility.
3. Graduates adopt new innovative technology by continuously upgrading their knowledge through lifelong learning for achieving personal and organization growth.

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI
B.E: Electronics & Communication Engineering / B.E: Electronics & Telecommunication Engineering NEP, Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2021 – 22)
V Semester

ELECTROMAGNETIC WAVES			
Course Code	21EC54	CIE Marks	50
Teaching Hours/Week (L: T: P: S)	(3:0:0:0)	SEE Marks	50
Total Hours of Pedagogy	40 hours Theory	Total Marks	100
Credits	03	Exam Hours	03

Course objectives: This course will enable students to :

- Study the different coordinate systems, Physical significance of Divergence, Curl and Gradient.
- Understand the applications of Coulomb's law and Gauss law to different charge distributions and the applications of Laplace's and Poisson's Equations to solve real time problems on capacitance of different charge distributions.
- Understand the physical significance of Biot-Savart's, Amperes's Law and Stokes'theorem for different current distributions.
- Infer the effects of magnetic forces, materials and inductance.
- Know the physical interpretation of Maxwell' equations and applications for Plane waves for their behavior in different media.
- Acquire knowledge of Poynting theorem and its application of power flow.

Teaching-Learning Process (General Instructions)

The sample strategies, which the teacher can use to accelerate the attainment of the various course outcomes are listed in the following:

1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes.
2. Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes criticalthinking
3. Adopt Problem Based Learning (PBL), which fosters students' analytical skills, develop thinkingskills such as the ability to evaluate, generalize & analyze information rather than simply recall it.
4. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.
5. Using videos for demonstration of the fundamental principles to students for better understandingof concepts.

Module-1

Revision of Vector Calculus – **(Text 1: Chapter 1)**

Coulomb's Law, Electric Field Intensity and Flux density: Experimental law of Coulomb, Electric field intensity, Field due to continuous volume charge distribution, Field of a line charge, Field due to Sheet of charge, Electric flux density, Numerical Problems. **(Text: Chapter 2.1 to 2.5, 3.1)**

Teaching-LearningProcess	Chalk and Talk would be helpful for the quantitative analysis. Videos of the Basicprinciples of the devices would help students to grasp better. **RBT Level:** L1, L2, L3
Module-2	
Gauss's law and Divergence: Gauss 'law, Application of Gauss' law to point charge, line charge, Surface charge and volume charge, Point (differential) form of Gauss law, Divergence. Maxwell's First equation (Electrostatics), Vector Operator ∇ and divergence theorem, Numerical Problems **(Text: Chapter 3.2 to 3.7).** **Energy, Potential and Conductors**: Energy expended or work done in moving a point charge in an electric field, The line integral, Definition of potential difference and potential, The potential field of point charge, Potential gradient, Numerical Problems **(Text: Chapter 4.1 to 4.4 and 4.6).**Current and Current density, Continuity of current. **(Text: Chapter 5.1, 5.2)**	
Teaching-Learning Process	Chalk and Talk, PowerPoint Presentation **RBT Level:** L1, L2, L3
Module-3	
Poisson's and Laplace's Equations: Derivation of Poisson's and Laplace's Equations, Uniqueness theorem, Examples of the solution of Laplace's equation, Numerical problems on Laplace equation **(Text: Chapter 7.1 to 7.3)** **Steady Magnetic Field**: Biot-Savart Law, Ampere's circuital law, Curl, Stokes' theorem, Magneticflux and magnetic flux density, Basic concepts Scalar and Vector Magnetic Potentials, Numerical problems. **(Text: Chapter 8.1 to 8.6)**	
Teaching-LearningProcess	Chalk and talk method, Power point presentation and videos. **RBT Level:** L1, L2, L3
Module-4	
Magnetic Forces: Force on a moving charge, differential current elements, Force between differential current elements, Numerical problems **(Text: Chapter 9.1 to 9.3).** **Magnetic Materials**: Magnetization and permeability, Magnetic boundary conditions, The magnetic circuit, Potential energy and forces on magnetic materials, Inductance and mutualreactance, Numerical problems **(Text: Chapter 9.6 to 9.7).** Faraday' law of Electromagnetic Induction –Integral form and Point form, Numerical problems **(Text: Chapter 10.1)**	
Teaching-LearningProcess	Chalk and Talk, PowerPoint Presentation **RBT Level:** L1, L2, L3
Module-5	
Maxwell's equations Continuity equation, Inconsistency of Ampere's law with continuity equation, displacement current, Conduction current, Derivation of Maxwell's equations in point form, and integral form, Maxwell's equations for different media, Numerical problems **(Text: Chapter 10.2 to 10.4)** **Uniform Plane Wave**: Plane wave, Uniform plane wave, Derivation of plane wave equations from Maxwell's equations, Solution of wave equation for perfect dielectric, Relation between E and H, Wave	

propagation in free space, Solution of wave equation for sinusoidal excitation, wave propagation in any conducting media (γ, α, β, η) and good conductors, Skin effect or Depth of penetration, Poynting's theorem and wave power, Numerical problems. **(Text: Chapter 12.1 to 12.4)**

Teaching-LearningProcess	Chalk and Talk, PowerPoint Presentation **RBT Level: L1, L2, L3**

Course Outcomes
At the end of the course the student will be able to:

- Evaluate problems on electrostatic force, electric field due to point, linear, volume charges by applyingconventional methods and charge in a volume.
- Apply Gauss law to evaluate Electric fields due to different charge distributions and Volume Charge distribution by using Divergence Theorem.
- Determine potential and energy with respect to point charge and capacitance using Laplace equation and Apply Biot-Savart's and Ampere's laws for evaluating Magnetic field for different currentconfigurations
- Calculate magnetic force, potential energy and Magnetization with respect to magnetic materials and voltage induced in electric circuits.
- Apply Maxwell's equations for time varying fields, EM waves in free space and conductors and Evaluate power associated with EM waves using Poynting theorem

Assessment Details (both CIE and SEE)
The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures not less than 35% (18 Marks out of 50) in the semester-end examination (SEE), and a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:
Three Unit Tests each of **20 Marks (duration 01 hour)**
1. First test at the end of 5th week of the semester
2. Second test at the end of the 10th week of the semester
3. Third test at the end of the 15th week of the semester

Two assignments each of **10 Marks**
4. First assignment at the end of 4th week of the semester
5. Second assignment at the end of 9th week of the semester

Group discussion/Seminar/quiz any one of three suitably planned to attain the COs and POs for **20 Marks (duration 01 hours)**
6. At the end of the 13th week of the semester

The sum of three tests, two assignments, and quiz/seminar/group discussion will be out of 100 marks **and will be scaled down to 50 marks**
(to have less stressed CIE, the portion of the syllabus should not be common /repeated for any of the methods of the CIE. Each method of CIE should have a different syllabus portion of the course).
CIE methods /question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester End Examination: Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the subject **(duration 03 hours)** 1. The question paper will have ten questions. Each question is set for 20 marks. 2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module. The students have to answer 5 full questions, selecting one full question from each module.
Suggested Learning Resources: **Text Book:** 1. W.H. Hayt and J.A. Buck, ―Engineering Electromagnetics‖, 8th Edition, Tata McGraw-Hill, 2014, ISBN-978-93-392-0327-6. **Reference Books:** 1. Elements of Electromagnetics – Matthew N.O., Sadiku, Oxford university press, 4th Edn. 2. Electromagnetic Waves and Radiating systems – E. C. Jordan and K.G. Balman, PHI, 2nd Edn. 3. Electromagnetics- Joseph Edminister, Schaum Outline Series, McGraw Hill. 4. N. NarayanaRao, ―Fundamentals of Electromagnetics for Engineering‖, Pearson
Web links and Video Lectures (e-Resources):
• https://archive.nptel.ac.in/courses/108/104/108104087/
Activity Based Learning (Suggested Activities in Class)/ Practical Based learning Quizzes, Seminars

ELECTROMAGNETIC THEORY

BECH01

MODULE - 01

Coulomb's law. Electric Field Intensity and Flux Density

Introduction:

Electrostatic deals with the cases in which electric charges are stationary, when the charges are stationary, the electric field set up by them remains static i.e., whenever magnitudes and direction of charges unchanged. Hence it is named as electrostatics.

Experimental law of coulomb:

Coulomb's law:

~~Ist~~ coulomb's law states that the electrostatic force F between two point charges Q_1 and Q_2 is directly proportional to the product of the magnitude of the charges. and inversely proportional to the square of the distance between the two charges. and it acts along the line joining the two charges.

Q_1 •—— r ——• Q_2

consider two point charges Q_1 and Q_2, Separated by a distance, r

- The Force 'F' between two charges Q_1 and Q_2 are Expressed as

$$F \propto \frac{Q_1 Q_2}{r^2}$$

or $$F = K \frac{Q_1 Q_2}{r^2} \quad \text{--- (1)}$$

where K = Proportionality constant

$Q_1 Q_2$ = Product of two charges

r = distance b/n two charges

- The value of Proportionality constant varies with medium where charges are located. If it is located in Free space then

$$K = \frac{1}{4\pi \varepsilon_0} \quad \text{--- (2)}$$ where ε_0 = Permittivity of free space.

- The Force can also be written as

$$F = \frac{1}{4\pi \varepsilon_0} \cdot \frac{Q_1 Q_2}{r^2} \quad N$$

- If $\hat{a}_r$ is the unit vector along the line joining the two charges, then

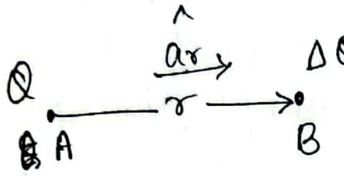
$$\boxed{F = \frac{Q_1 Q_2}{4\pi \varepsilon_0 r^2} \hat{a}_r} \quad \text{--- (3)}$$

- If the medium is not Free space then

$$F = \frac{Q_1 Q_2}{4\pi \varepsilon r^2} \hat{a}_r$$ where $\varepsilon = \varepsilon_0 \varepsilon_r$ — relative permittivity.

Electric Field Intensity:

defⁿ: Electric Field intensity at any point in an electric field is the force experienced by a unit positive charge placed at ~~any~~ that point.



Fig① charge

- Consider a charge 'Q' located at a point 'A'. At the point B in the electric field ~~intensity~~ setup by Q, it is required to find electric field intensity E.

- According to coloumb's Law, the force between Q and ΔQ is given by

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{Q(\Delta Q)}{r^2} \quad — ①$$

- If ΔQ is the unit charge and ΔQ=1, then force given the ~~....~~ electric field intensity, E. E=F.

$$\therefore \quad E = \frac{Q}{4\pi\varepsilon_0 r^2} \quad — ②$$

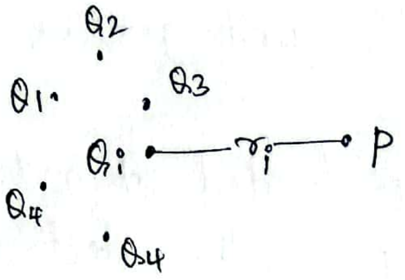
- vector representation of EFI is

$\hat{a}_r$ – unit vector

$$\boxed{\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2} \hat{a}_r} \quad — ③$$

This gives Electric field intensity at a point due to point charge.

Electric Field at a point due to many charges:



Note: The principal of superposition states that, the force between any two charges is independent of the presence of other charges, due to which the effect of individual charges would be superposed.

- The field at a point due to two or more charges is the vector sum of the fields that the various charges would setup individually at the same point.
- $Q_1, Q_2 \dots Q_n$ are many charges, at a distance $r_1, r_2, \dots r_n$ from the point 'P'. Let $\hat{a}_{r_1}, \hat{a}_{r_2} \dots \hat{a}_{r_n}$ are the corresponding unit vectors.
- The Electric Field intensity at a point 'P' due to the charges, $Q_1, Q_2 \dots Q_n$ are respectively as follows

$$\bar{E}_1 = \frac{Q_1}{4\pi\varepsilon_0 r_1^2}\hat{a}_{r_1}, \quad \bar{E}_2 = \frac{Q_2}{4\pi\varepsilon_0 r_2^2}\hat{a}_{r_2} \dots\dots \bar{E}_n = \frac{Q_n}{4\pi\varepsilon_0 r_n^2}$$

- The total Field $\bar{E}$ at P due to all the charges can be written by using the principle of superposition as,

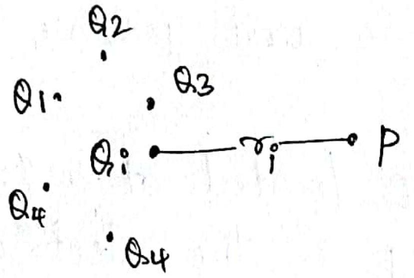
$$\bar{E} = \bar{E}_1 + \bar{E}_2 + \dots \bar{E}_n$$

$$\bar{E} = \frac{Q_1}{4\pi\varepsilon_0 r_1^2}\hat{a}_{r_1} + \frac{Q_2}{4\pi\varepsilon_0 r_2^2}\hat{a}_{r_2} + \dots\dots + \frac{Q_n}{4\pi\varepsilon_0 r_n^2}\hat{a}_{r_n}$$

$$\therefore \boxed{\bar{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i^2}\hat{a}_{r_i}}$$ ⇒ EFI due to many charges.

M-01.

Electric Field at a point due to many charges:



Note: The principal of Superposition states that, the force between any two charges is independent of the presence of other charges, due to which the effect of individual charges would be superposed.

- The field at a point due to two or more charges is the vector sum of the fields that the various charges would setup individually at the same point.
- $Q_1, Q_2 \cdots Q_n$ are many charges, at a distance $r_1, r_2, \cdots r_n$ from the point 'P'. Let $\hat{a}_{r_1}, \hat{a}_{r_2} \cdots \hat{a}_{r_n}$ are the corresponding unit vectors.
- The Electric Field intensity at a point 'P' due to the charges $Q_1, Q_2 \cdots Q_n$ are respectively as follows

$$\bar{E}_1 = \frac{Q_1}{4\pi\varepsilon_0 r_1^2}\hat{a}_{r_1}, \quad \bar{E}_2 = \frac{Q_2}{4\pi\varepsilon_0 r_2^2}\hat{a}_{r_2} \cdots\cdots \bar{E}_n = \frac{Q_n}{4\pi\varepsilon_0 r_n^2}$$

- The total Field $\bar{E}$ at P due to all the charges can be written by using the principle of Superposition as,

$$\bar{E} = \bar{E}_1 + \bar{E}_2 + \cdots \bar{E}_n$$

$$\bar{E} = \frac{Q_1}{4\pi\varepsilon_0 r_1^2}\hat{a}_{r_1} + \frac{Q_2}{4\pi\varepsilon_0 r_2^2}\hat{a}_{r_2} + \cdots\cdots + \frac{Q_n}{4\pi\varepsilon_0 r_n^2}\hat{a}_{r_n}$$

$$\therefore \boxed{\bar{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i^2}\hat{a}_{r_i}} \Rightarrow \text{EFI due to many charges.}$$

Electric Field Intensity at a point due to infinite number of point charges:

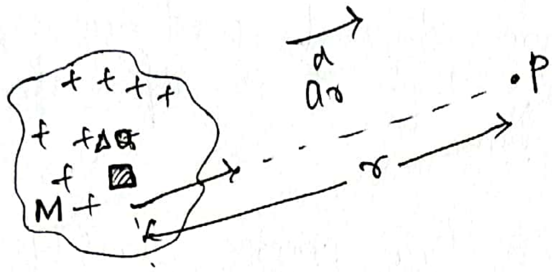


Fig: Infinite charges over a space

Consider infinite charges over elementary space 'M'. In that consider a small amount of charge 'ΔQ' and 'p' is the point where EFI is determined.

'r' is the distance b/n 'ΔQ' and 'P'. $\hat{a}_r$ is unit vector.

- Electric Field intensity at 'P' due to charge ΔQ is,

$$\Delta\bar{E} = \frac{\Delta Q}{4\pi\epsilon_0 r^2}\hat{a}_r \quad —①$$

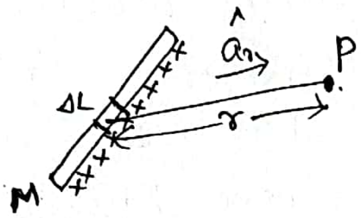
- Let $\Delta\bar{E} = d\bar{E}$ & $\Delta Q = dQ$ then rewrite above Eqn

$$d\bar{E} = \frac{dQ}{4\pi\epsilon_0 r^2}\hat{a}_r \quad —②$$

∴ The EFI at 'P' due to the entire charge distribution. i.e., (all infinite charge) is given by integrating Eqn②.

$$\boxed{E = \int d\bar{E} = \int \frac{dQ}{4\pi\epsilon_0 r^2}\hat{a}_r} \quad —③$$

Electric Field Intensity at a point due to line charge distribution:



Eg: Infinite line Charge distribution

- If the charge distribution is such that the charges are aligned continuously along a line. then it is referred as line charge.

- let ρ_L be the line charge density and is defined as the charge per unit length.

i.e., $\rho_L = \left[\dfrac{\Delta Q}{\Delta L}\right]_{\Delta L \to 0}$ or $\rho_L = \dfrac{dQ}{dL} \left(\dfrac{C}{m}\right)$ — (1)

Note.
$\{dQ = \rho_L \, dL\}$.

W.r.t the EFI at point due to line charge is.

$$\bar{E} = \int \frac{dQ}{4\pi\varepsilon_0 r^2}\hat{a}_r = \frac{1}{4\pi\varepsilon_0}\int \frac{dQ}{r^2}\hat{a}_r \quad \text{—(2)}$$

Using $dQ = \rho_L \, dL$ in Eqn (2)

$$\bar{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho_L \, dL}{r^2}\hat{a}_r$$

or

$$\boxed{\bar{E} = \frac{\rho_L}{4\pi\varepsilon_0}\int \frac{dL}{r^2}\hat{a}_r} \quad \text{—(3)}$$

is the EFI due to infinite line charge.

Electric Field Intensity at a point due to a Sheet of charge:

- If the charge distribution is such that the charges are continuously distributed on a two dimensional surface, then it is referred to as "Surface charge distribution" or "Sheet of charge".

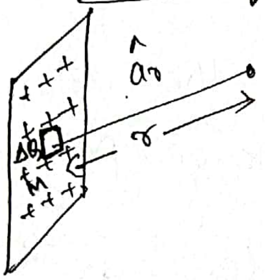


Fig: Surface charge distribution.

- consider a surface 'S' where infinite charges distributed. let ρ_s be the surface charge density, defined as the charge per - unit surface area.

i.e., $\rho_s = \frac{\Delta Q}{\Delta S}_{\Delta S \to 0}$ (or) $\rho_s = \frac{dQ}{ds} \frac{C}{m^2}$ —(1)

W.K.T. The EFI at differential form. i.e.,

$$d\bar{E} = \frac{dQ}{4\pi\epsilon_0 r^2}\hat{a}_r$$

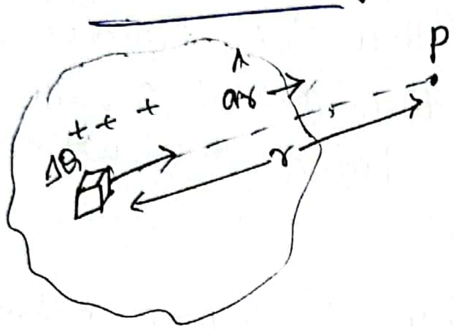
Since the charge distribution is continuous, the field at P due to the entire charged sheet is.

$$\bar{E} = \int d\vec{E} = \int \frac{dQ}{4\pi\epsilon_0 r^2}\hat{a}_r = \frac{1}{4\pi\epsilon_0}\int \frac{dQ}{r^2}\hat{a}_r$$

$$= \frac{1}{4\pi\epsilon_0}\int \frac{\rho_s ds}{r^2} a_r \qquad \{\text{using } dQ = \rho_s ds\}$$

or $\boxed{\bar{E} = \frac{\rho_s}{4\pi\epsilon_0}\int \frac{ds}{r^2}\hat{a}_r}$ given EFI due to infinite Sheet of charge.

Electric Field Intensity at a point due to Continuous Volume charge Distribution:



If the charge distribution is such that the charges are distributed continuously in a volume, then it is referred as a volume charge distribution.

- For the volume charge distribution, the volume charge density ρ_v can be defined as the charge per unit volume.

i.e., $\rho_v = \frac{\Delta Q}{\Delta v}_{\Delta v \to 0}$ (or) $\rho_v = \frac{dQ}{dv}$ $\frac{C}{m^3}$ — (1)

W.K.T $\quad d\bar{E} = \frac{dQ}{4\pi\varepsilon_0 r^2}\hat{a}_r$ — (2)

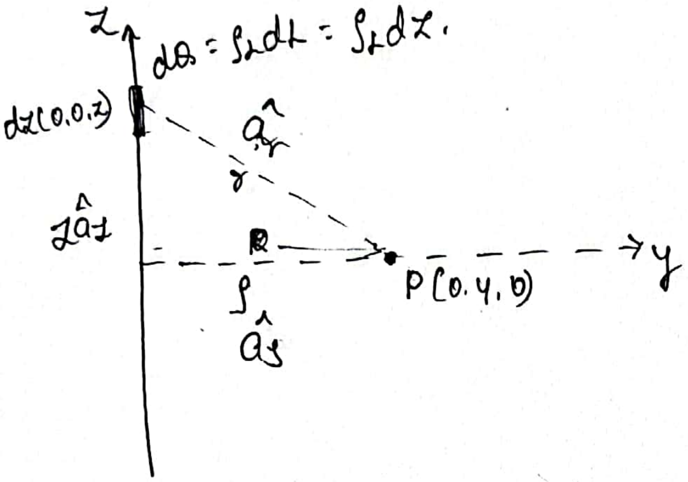
Since charge distribution is contineous, the field at 'p' due to the entire volume charge distribution is.

$$\vec{E} = \int d\bar{E} = \int \frac{dQ}{4\pi\varepsilon_0 r^2}\hat{a}_r = \int \frac{\rho_v\, dv}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

$$\boxed{\bar{E} = \frac{\rho_v}{4\pi\varepsilon_0}\int_v \frac{dv}{r^2}\hat{a}_r}$$ $\frac{V}{m}$ (or) $\frac{N}{C}$ — (3)

is the Electric Field Intensity due to Volume Charge distribution.

Electric Field due to a line charge (Uniformly charged wire):



- Consider a uniform charge configuration on z-axis. (i.e.. Source point).
- It can be compared with cylindrical co-ordinate System. because it has only z-axis. as source point
  $\therefore$ P (ρ, ϕ, z).

- Electric field intensity changes with change in 'ρ'. Variation of 'ϕ' and 'z' will not make any changes on $\vec{E}$.
- Consider charges of differential length i.e dQ on z axis.
  w.k.T line charge density $\rho_L = \frac{dQ}{dL}$ (A) $dQ = \rho_L dL$
  (A) $dQ = \rho_L dz$. —(1)

- Electric Field intensity due to point charge

$$E = \frac{Q}{4\pi\epsilon_0 r^2}\hat{a}_r \quad \frac{V}{m}$$

$$\bar{dE} = \frac{dQ}{4\pi\epsilon_0 r^2}\hat{a}_r \qquad \{ \text{ differentiation of } \bar{E}$$

$$\bar{E} = \int_{z=-\infty}^{\infty} dE = \int_{z=-\infty}^{\infty} \frac{\rho_L\, dz}{4\pi\epsilon_0 r^2}\hat{a}_r \quad \text{—(2)}$$

from fig: From Δ^{le} law of addition

$$z\hat{a}_z + r\hat{a}_r = \rho\hat{a}_\rho$$

$$r\hat{a}_r = \rho\hat{a}_\rho - z\hat{a}_z$$

$$\hat{a}_r = \frac{\rho\hat{a}_\rho - z\hat{a}_z}{r}$$

From Rt angle triangle

$$r = \sqrt{\rho^2 + z^2}$$

$$\therefore \hat{a}_r = \frac{\rho\hat{a}_\rho - z\hat{a}_z}{\sqrt{\rho^2+z^2}} \quad \text{(3)}$$

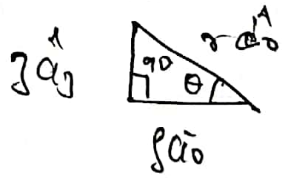
Using Eqn (3) in (2)

$$\bar{E} = \int_{z=-\infty}^{\infty} \frac{\rho_L \, dz}{4\pi\epsilon_0 \left(\sqrt{\rho^2+z^2}\right)^2} \cdot \frac{\rho\hat{a}_\rho - z\hat{a}_z}{\sqrt{\rho^2+z^2}}$$

neglect $z\hat{a}_z$ term $\because$ variation in ϕ and z will not affect $\vec{E}$.

$$\therefore \bar{E} = \int_{z=-\infty}^{\infty} \frac{\rho_L}{4\pi\epsilon_0 (\rho^2+z^2)^{2/3}} \rho\hat{a}_\rho \, dz$$

From



$\tan\theta = \frac{z}{\rho}$

$z = \rho\tan\theta$

$$= \int_{\theta=-\pi/2}^{\pi/2} \frac{\rho_L \, \rho \, (\rho\sec^2\theta)}{4\pi\epsilon_0 (\rho^2 + \rho^2\tan^2\theta)^{2/3}} d\theta \cdot \hat{a}_\rho$$

$\theta = \tan^{-1}(z)$ $\quad dz = \rho\sec^2\theta \, d\theta$

$\tan^{-1}\infty = \pi/2$.

$z = -\infty \quad \theta = -\frac{\pi}{2}$

$z = \infty \quad \theta = \frac{\pi}{2}$

M1-10

$$= \int_{\theta=-\pi/2}^{\pi/2} \frac{\rho_L}{4\pi\varepsilon_0 \rho (1+\tan^2\theta)^{3/2}} \cdot \sec^2\theta \, d\theta \, \hat{a}_y$$

$$\int \frac{\rho_L \rho^2 \sec^2\theta}{4\pi\varepsilon_0 \rho^{3/2 \times 2} (1+\tan^2\theta)^{3/2}} \, d\theta$$

$$\frac{\rho^2}{\rho^{3/2 \times 2}} = \frac{1}{\rho}$$

$$\frac{\sec^2\theta}{(1+\tan^2\theta)^{3/2}}$$

$$(\sec^2\theta)^{3/2}$$

$$\frac{1}{\sec\theta}$$

$$= \int_{\theta=-\pi/2}^{\pi/2} \frac{\rho_L}{4\pi\varepsilon_0 \rho} \frac{1}{\sec\theta} \, d\theta \, \hat{a}_y$$

$$= \frac{\rho_L}{4\pi\varepsilon_0 \rho} \int_{\theta=-\pi/2}^{\pi/2} \cos\theta \, d\theta \, \hat{a}_y$$

$$= \frac{\rho_L}{4\pi\varepsilon_0 \rho} (\sin\theta)_{\theta=-\pi/2}^{\pi/2} \hat{a}_y$$

$$= \frac{\rho_L}{4\pi\varepsilon_0 \rho} [1-(-1)] \hat{a}_y$$

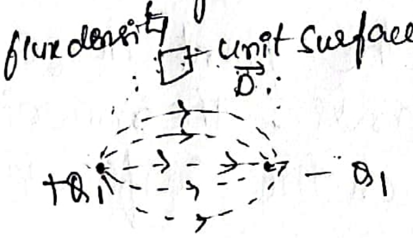
$$\frac{1}{2}+1$$

$$\frac{3}{2}$$

$$\therefore \boxed{\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0 \rho} \hat{a}_y} \ \text{V/m}.$$

Electric Flux Density:

- Consider the two point charges as shown in figure ①
The flux lines originating from +ve charge and terminating at negative charge are



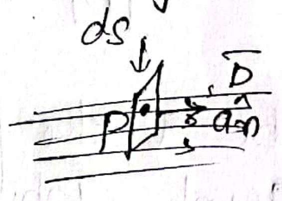
consider unit surface area (S), the net flux (flux lines) passing through that area is called Electric flux density ⓓ denoted by 'D'

The Electric flux density is ~~the~~ defined as total flux over surface area.

i.e $D = \frac{\psi}{S}$ where ψ = Total flux
S = Total surface area

The vector form of Electric flux density D is

$$\vec{D} = \frac{d\psi}{ds}\,\hat{a}_n \quad C/m^2$$



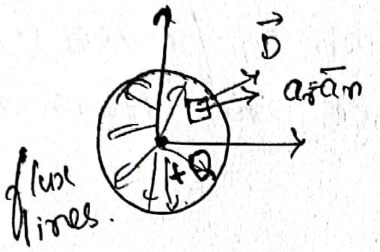
where $d\psi$ = ~~Total~~ flux lines passing normal through the ds differential area

ds = differential surface area

$\hat{a}_n$ = unit vector normal to differential surface area.

M1 — a

Electric flux density due to point charge Q.



- Consider a point charge +Q placed at the centre of the imaginary sphere of radius 'r'.

- The flux lines originated from the point charge +Q are directed radially outwards. The magnitude of the flux density at any point on the surface is.

$$D = \frac{\text{Total flux } \psi}{\text{Total surface area } S} \quad \text{——} (1) \quad \frac{C}{m^2}$$

where $\psi = Q$ = Total flux

$S = 4\pi r^2$ Total surface area.

$$|D| = \frac{Q}{4\pi r^2}$$

- The vector form of Electric flux density 'D' at a point which is at a distance of r, from the point charge +Q is given by.

$$\boxed{\vec{D} = \frac{Q}{4\pi r^2}\,\hat{a}_r \quad C/m^2} \quad \text{——} (2).$$

⇒ Relationship Between $\bar{D}$ and $\bar{E}$:

- Electric field intensity due to point charge is

$$\bar{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{a}_r$$

- The ratio of D by E is

$$\frac{\bar{D}}{\bar{E}} = \frac{\frac{Q}{4\pi r^2}\hat{a}_r}{\frac{Q}{4\pi\varepsilon_0 r^2}\hat{a}_r} = \varepsilon_0$$

$$\therefore \boxed{\bar{D} = \varepsilon_0 \bar{E}}$$

$D = \varepsilon \bar{E}$

$D = \varepsilon_0 \varepsilon_r \bar{E}$

$D = \varepsilon_0 \bar{E}$ where $\varepsilon_r = 1$ for free space.

Note:- (1) $\bar{E}$ and $\bar{D}$ acts in same direction

(2) $\bar{D}$ and E are related through the permittivity of the medium in which charge is located.

(3) $\bar{E}$ due to any charge configuration is a function of permittivity ε, while $\vec{D}$ is not.

Electric flux density for various charge Distribution:

(1) **Electric flux density due to Line charge:**

Let ρ_L - line charge density. C/m $\rho_L = \frac{dQ}{dL}$ $dQ = \rho_L dL$.

The total charge along line is,

$$Q = \int_L \rho_L \, dl$$

But $D = \frac{Q}{4\pi r^2}\hat{a}_r = \frac{\int \rho_L \, dL}{4\pi r^2}\hat{a}_r$

$$D = \frac{1}{4\pi r^2} \int \rho_L \, dL \, \hat{a}_r$$

If line charge is infinite then E in

$$E = \frac{\rho_L}{2\pi \varepsilon_0 r} \hat{a}_r$$

and $D = \varepsilon_0 E$

$$\therefore \boxed{D = \frac{\rho_L}{2\pi r} \hat{a}_r}$$ — Electric flux density due to infinite line charge.

(2) Electric flux density due to infinite surface charge:

consider a sheet of charge having uniform charge density ρ_s C/m^2.

– Electric field intensity due to infinite sheet of charge in

$$\bar{E} = \frac{\rho_s}{2\varepsilon_0} \hat{a}_n \quad V/m$$

and $\bar{D} = \varepsilon_0 \bar{E}$

$$\therefore \boxed{\bar{D} = \frac{\rho_s}{2} \hat{a}_n}$$

2-d

2-f

③ Electric flux density due to infinite volume charge:

- Consider uniform volume charge density ρ_v C/m³.
- The electric field intensity due to infinite volume charge is.

$$\bar{E} = \frac{1}{4\pi\varepsilon_0 r^2}\int_v \rho_v\, dv \; \hat{a}_r$$

and $\bar{D} = \varepsilon_0 \bar{E}$

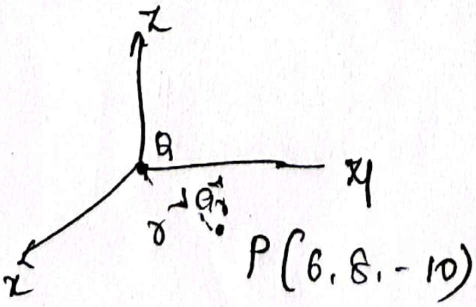
$$\therefore \boxed{\bar{D} = \frac{1}{4\pi r^2}\int_v \rho_v\, dv \; \hat{a}_r}$$

Electric flux density due to infinite volume charge.

Problem:

① Find D in cartesian co-ordinate system at a point P(6, 8, −10) due to (a) point charge of 40 mC at the origin (b) a uniform line charge of ρ_L = 40 μC/m on the x-axis and (c) a uniform surface charge of density ρ_s = 57.2 μC/m² on the plane x = 12 m.

Soln:



2-e

(a) $\vec{r} = (6-0)\hat{a}_x + (8-0)\hat{a}_y + (-10-0)\hat{a}_z$

$\vec{r} = 6\hat{a}_x + 8\hat{a}_y - 10\hat{a}_z$

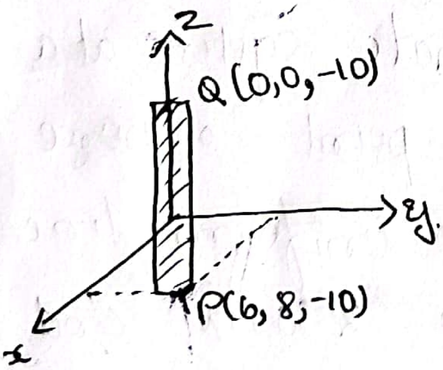
$|\vec{r}| = \sqrt{200}$

$$\hat{a}_r = \frac{6\hat{a}_x + 8\hat{a}_y - 10\hat{a}_z}{\sqrt{200}}$$

$$\bar{D} = \frac{Q}{4\pi r^2}\hat{a}_r = \frac{40\times 10^{-3}}{4\pi(200)} \times \frac{6\hat{a}_x + 8\hat{a}_y - 10\hat{a}_z}{\sqrt{200}}$$

$$\bar{D} = 6.752\times 10^{-6}\hat{a}_x + 9.003\times 10^{-6}\hat{a}_y - 11.254\times 10^{-6}\hat{a}_z \ C/m^2$$

(b) $\rho_L = 40\,\mu C/m$ along z axis $(0, 0, -10)$



$$\bar{D} = \frac{\rho_L}{2\pi\rho}\hat{a}_\rho$$

$\bar{\rho} = (6-0)\hat{a}_x + (8-0)\hat{a}_y + (-10-(-10))\hat{a}_z$

$\bar{\rho} = 6\hat{a}_x + 8\hat{a}_y$

$|\rho| = \sqrt{6^2+8^2} = \sqrt{100}$

$$\hat{a}_\rho = \frac{6\hat{a}_x + 8\hat{a}_y}{\sqrt{100}}$$

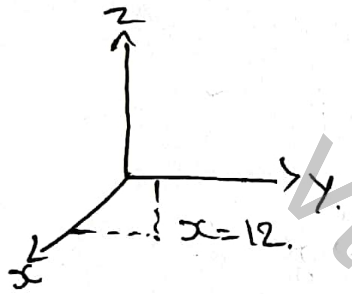
$$\vec{D} = \frac{\rho_L}{2\pi\rho}\hat{a}_\rho$$

$$= \frac{40 \times 10^{-6}}{2 \times \pi \times \sqrt{100}} \cdot \frac{6\hat{a}_x + 8\hat{a}_y}{\sqrt{100}}$$

$$= 6.36 \times 10^{-8} (6\hat{a}_x + 8\hat{a}_y)$$

$$\bar{D} = 3.816 \times 10^{-7} \hat{a}_x + 5.088 \times 10^{-7} \hat{a}_y$$

c) uniform surface of density $\rho_s = 57\mu C/m^2$ on the plane $x = 12m$.

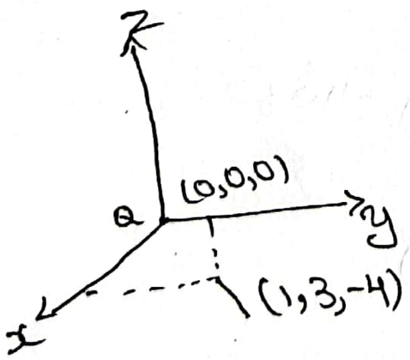


$$\bar{D} = \frac{\rho_s}{2} \hat{a}_n.$$

$$= \frac{57.2 \times 10^{-6}}{2} \times \hat{a}_x$$

$$\bar{D} = 2.86 \times 10^{-05} \hat{a}_x.$$

② A point charge $Q = 30nC$ is located at the origin in cartesian. co-ordinates. Find the electric flux density and electric field intensity at $(1, 3, -4)$ m.



Electric field intensity

$$\vec{E} = \frac{Q}{4\pi \epsilon_0 r^2} \hat{a}_r.$$

$$\vec{r} = (1-0)\hat{a}_x + (3-0)\hat{a}_y + (-4-0)\hat{a}_z$$

$$\vec{r} = 1\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z.$$

$$|\vec{r}| = \sqrt{(1^2) + (3)^2 + (-4)^2} = \sqrt{26}$$

$$\hat{a}_r = \frac{\vec{r}}{|\vec{r}|}$$

$$\hat{a}_r = \frac{1\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z}{\sqrt{26}}.$$

$$\vec{E} = \frac{30 \times 10^{-9}}{4 \times 3.14 \times (8.854 \times 10^{-12}) \times 26} \; \frac{\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z}{\sqrt{26}}$$

$$\vec{E} = 2.034 \, (\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z)$$

$$\vec{E} = 2.034\,\hat{a}_x + 6.102\,\hat{a}_y - 8.136\,\hat{a}_z.$$

Electric ~~field~~ flux intensity

$$\vec{D} = \frac{Q}{4\pi r^2}\,\hat{a}_r.$$

$$\vec{r} = (1-0)\hat{a}_x + (3-0)\hat{a}_y + (-4-0)\hat{a}_z$$

$$\vec{r} = \hat{a}_x + 3\hat{a}_y - 4\hat{a}_z.$$

$$|\vec{r}| = \sqrt{(1^2) + (3)^2 + (-4)^2}$$

$$|\vec{r}| = \sqrt{26}.$$

$$\hat{a}_r = \frac{\vec{r}}{|\vec{r}|}$$

$$\hat{a}_r = \frac{1\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z}{\sqrt{26}}.$$

$$\vec{D} = \frac{30 \times 10^{-9}}{4 \times 3.14 \times 26} \; \frac{1\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z}{\sqrt{26}}$$

$$\vec{D} = 1.80 \times 10^{-11} \, (1\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z)$$

$$\vec{D} = 1.80 \times 10^{-11}\hat{a}_x + 5.4 \times 10^{-11}\hat{a}_y - 7.2 \times 10^{-11}\hat{a}_z.$$

2-h.

③ A point charge of 6 μC is located at the origin a uniform line charge density 180 nC/m lies along x axis and uniform sheet charge of charge equal to 25 nC/m² lies in the z-plane.

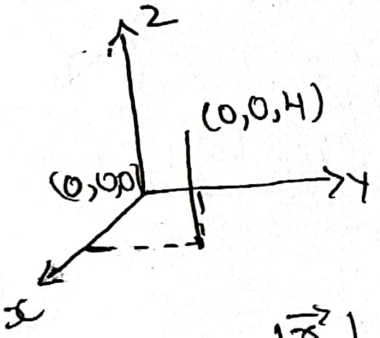
Find.

i) $\bar{D}$ at A (0,0,4)

ii) $\bar{D}$ at B (1,2,4).

i) $\vec{D}$ at A (0,0,4).

case i):- Point charge of 6 μC.



$$\vec{D} = \frac{Q}{4\pi r^2}\hat{a}_r$$

$$\vec{r} = (0-0)\hat{a}_x + (0-0)\hat{a}_y + (4-0)\hat{a}_z$$

$$\vec{r} = 4\hat{a}_z$$

$$|\vec{r}| = \sqrt{(4)^2} = \sqrt{16} = 4.$$

$$\hat{a}_r = \frac{\vec{r}}{|\vec{r}|}$$

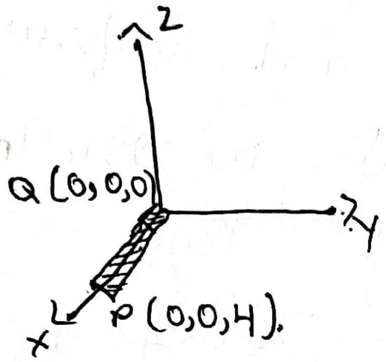
$$\hat{a}_r = \frac{4\hat{a}_z}{4}.$$

$$\vec{D} = \frac{6\times10^{-6}}{4\times3.14\times(4)^2}\ \frac{4\hat{a}_z}{4}.$$

$$\vec{D} = 7.460\times10^{-9}\ (4\hat{a}_z)$$

$$\vec{D} = 2.98\times10^{-8}\ \hat{a}_z$$

case 2). uniform line charge density



$$\vec{D} = \frac{\rho_L}{2\pi\rho}\hat{a}_\rho$$

$$\vec{\rho} = (0-0)\hat{a}_x + (0-0)\hat{a}_y + (4-0)\hat{a}_z$$

$$\vec{\rho} = 4\hat{a}_z$$

$$|\vec{\rho}| = \sqrt{(4^2)} = \sqrt{16} = 4.$$

$$\hat{a}_\rho = \frac{\vec{\rho}}{|\vec{\rho}|}$$

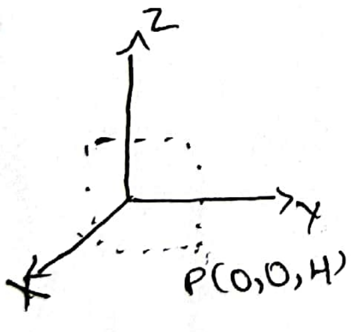
$$\hat{a}_\rho = \frac{4\hat{a}_z}{4}$$

$$\vec{D} = \frac{180\times10^{-9}}{2\times3.14\times4}\ \frac{4\hat{a}_z}{4.}$$

$$\vec{D} = 1.79\times10^{-9}\times4\hat{a}_z$$

$$\vec{D} = 7.16\times10^{-9}\hat{a}_z$$

case 3 :- uniform sheet charge.



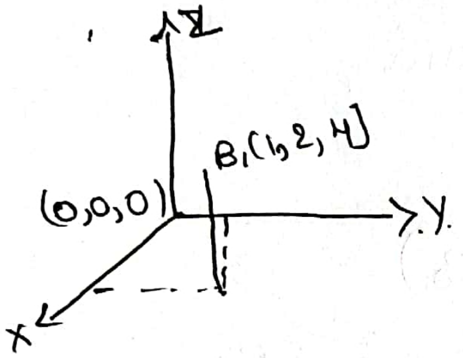
$$\rho_s = \frac{\rho}{2}\hat{a}_z$$

$$\rho_s = \frac{25\times10^{-9}}{2}\hat{a}_z$$

$$\rho_s = 1.25\times10^{-8}\hat{a}_z$$

ii) $\vec{D}$ at $B(1,2,4)$.

case i) Electric flux density at a point charge



$$\vec{D} = \frac{Q}{4\pi r^2}\hat{a}_r$$

$$\vec{r} = (1-0)\hat{a}_x + (2-0)\hat{a}_y + (4-0)\hat{a}_z$$

$$\vec{r} = 1\hat{a}_x + 2\hat{a}_y + 4\hat{a}_z.$$

$$|\vec{r}| = \sqrt{(1^2)+(2^2)+(4^2)}$$

$$|\vec{r}| = \sqrt{21}$$

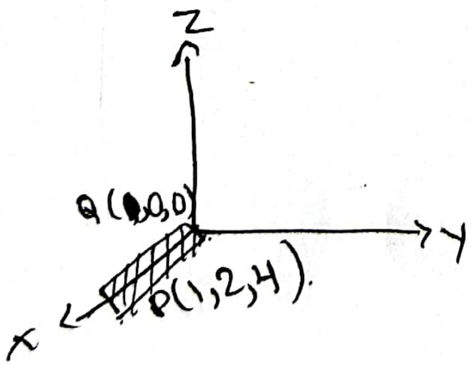
$$\hat{a}_r = \frac{\vec{r}}{|\vec{r}|} = \frac{\hat{a}_x + 2\hat{a}_y + 4\hat{a}_z}{\sqrt{21}}$$

$$\vec{D} = \frac{6\times10^{-6}}{4\times3.14\times21}\;\frac{\hat{a}_x + 2\hat{a}_y + 4\hat{a}_z}{\sqrt{21}}.$$

$$\vec{D} = 4.96\times10^{-9}(\hat{a}_x + 2\hat{a}_y + 4\hat{a}_z)$$

$$\vec{D} = 4.96\times10^{-9}\hat{a}_x + 9.92\times10^{-9}\hat{a}_y + 19.84\times10^{-9}\hat{a}_z.$$

case ii) uniform line charge.



$$\vec{D} = \frac{\rho_L}{2\pi\rho}\hat{a}_\rho.$$

$$\vec{\rho} = (1-1)\hat{a}_x + (2-0)\hat{a}_y + (4-0)\hat{a}_z$$

$$\vec{\rho} = 2\hat{a}_y + 4\hat{a}_z.$$

$$|\vec{\rho}| = \sqrt{(2^2)+(4^2)} = \sqrt{20}.$$

$$\hat{a}_\rho = \frac{\vec{r}}{|\vec{\rho}|}$$

2-K.

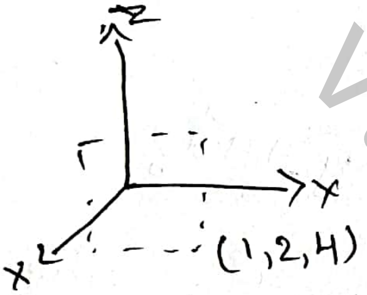
$$\hat{a}_\rho = \frac{2\hat{a}_y + 4\hat{a}_z}{\sqrt{20}}$$

$$\vec{D} = \frac{180 \times 10^{-9}}{2 \times 3.14 \times \sqrt{20}} \; \frac{2\hat{a}_y + 4\hat{a}_z}{\sqrt{20}}$$

$$\vec{D} = 1.434 \times 10^{-9} \left(2\hat{a}_y + 4\hat{a}_z\right)$$

$$\vec{D} = 2.868 \times 10^{-9}\,\hat{a}_y + 5.736 \times 10^{-9}\,\hat{a}_z$$

case iii) uniform sheet charge.



$$\vec{D} = \frac{\rho_s}{2}\,\hat{a}_s$$

$$\vec{D} = \frac{25 \times 10^{-9}}{2}$$

$$\vec{D} = 1.25 \times 10^{-8}\,\hat{a}_z$$